How Helsinki Uses GIS and AI to Shape Urban Development

Viva Amanifard

Urban planner

Oct 2026

Helsinki



Content

- Urban planning system
- GIS in urban planning and decision-making
- Urban planning and AI in practice



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Finnish Planning System



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Population of Helsinki

City of Helsinki 2025 (Dec)

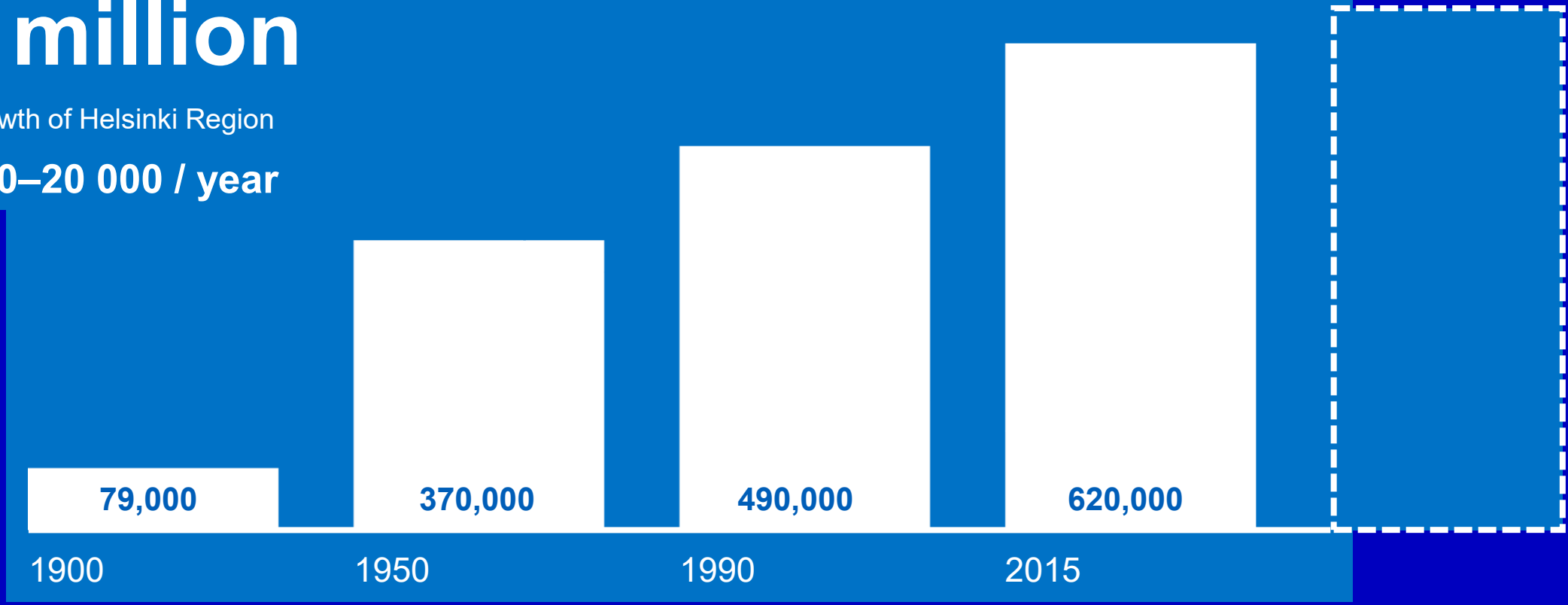
694 392

Total of Helsinki region

1.6 million

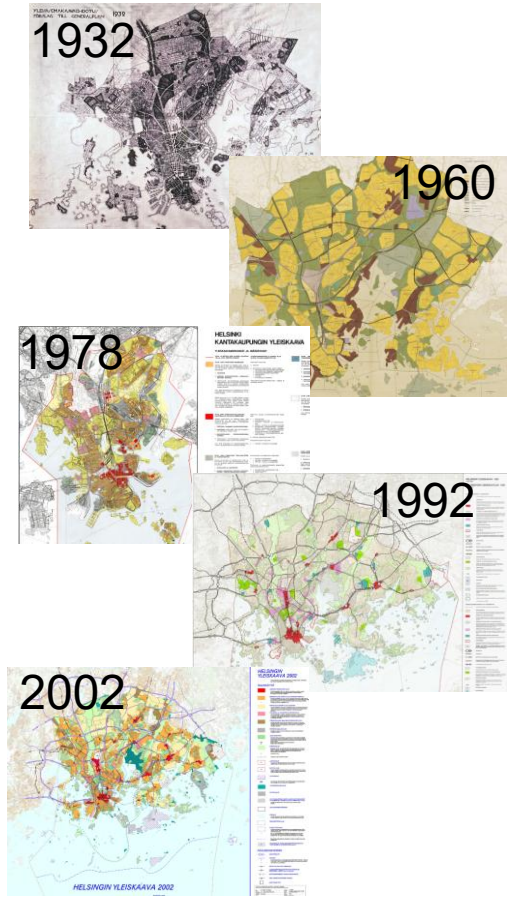
Annual growth of Helsinki Region

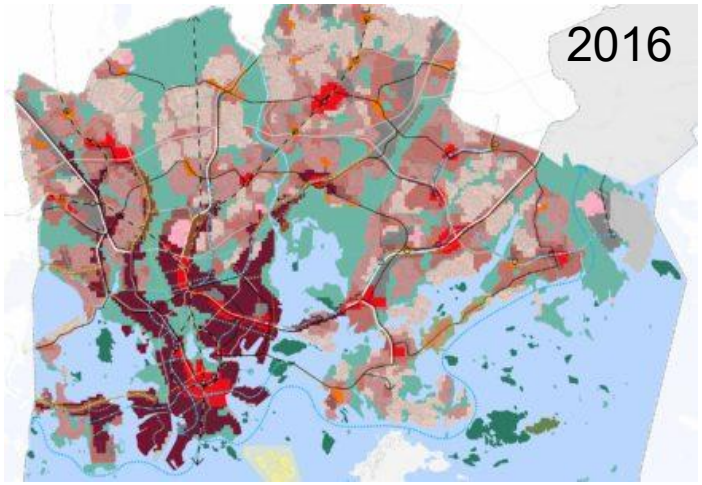
+ 15 000–20 000 / year



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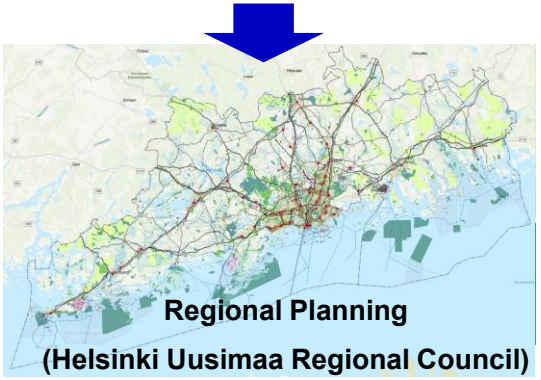
Finnish Planning System

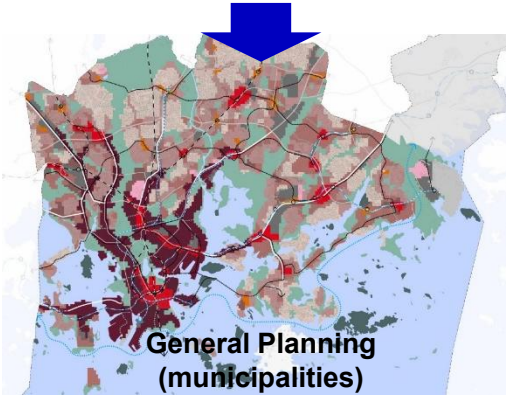


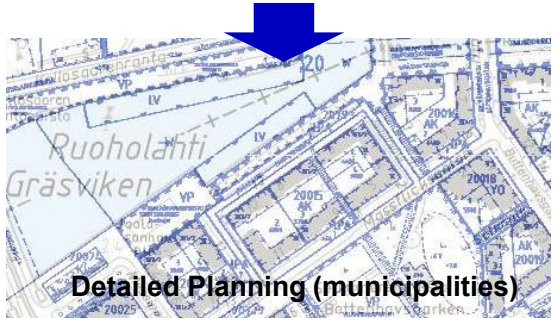


1. Expansion of the inner city
2. Creating a Public Transport Network City
3. Infill development of important nodes
4. Improving the green network city
5. Carbon neutrality

Land use and Building Act
Construction Act
Etc...





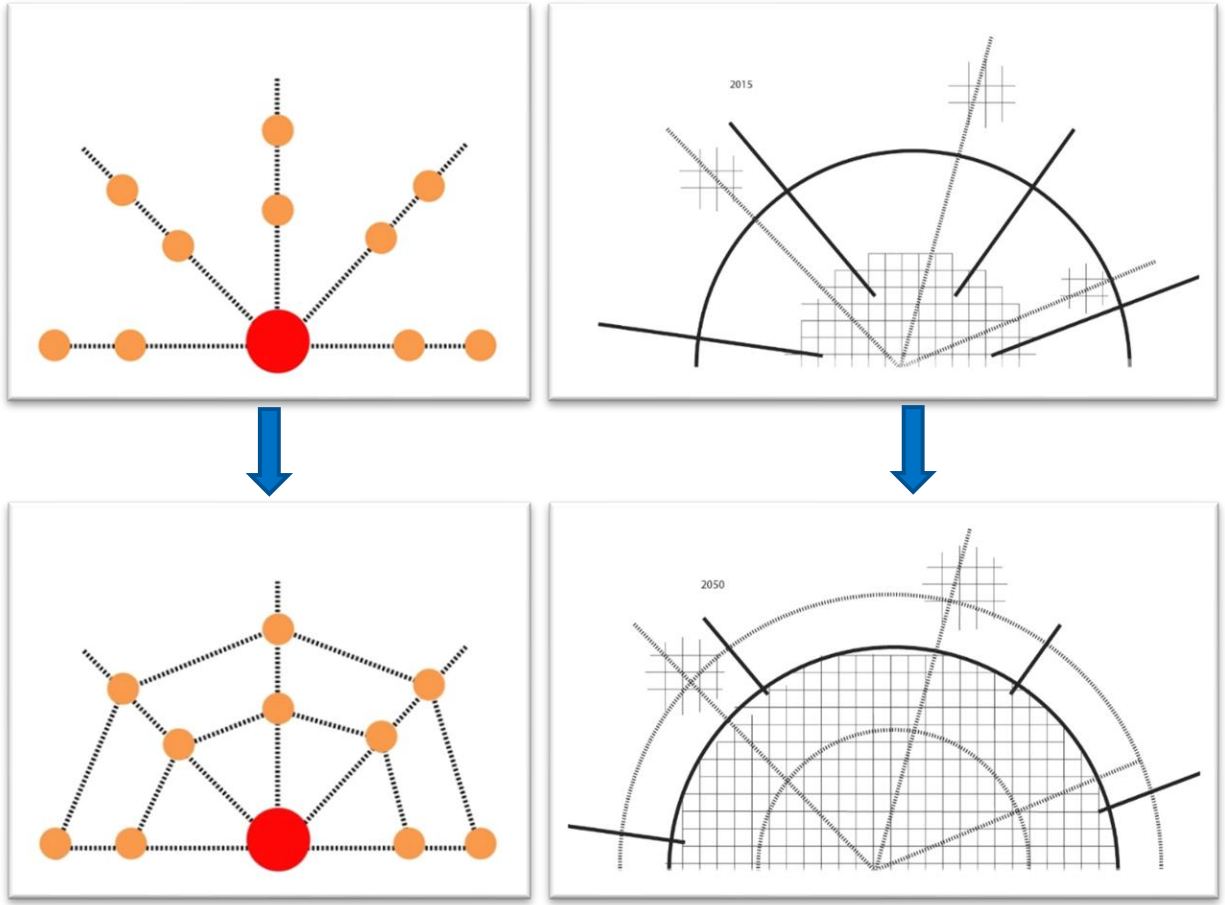


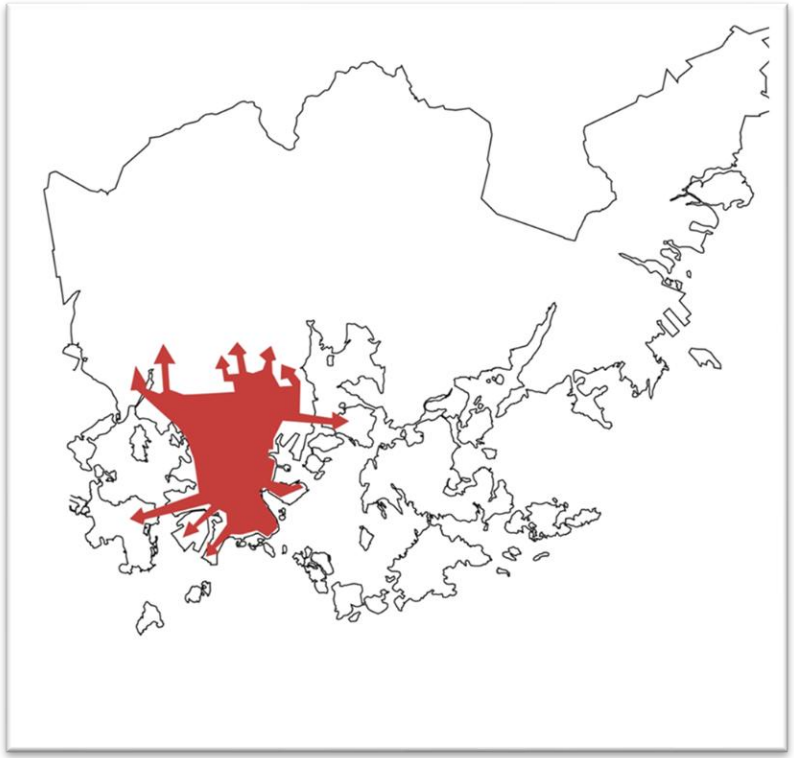
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Transforming the urban structure

Public transportation

Land use





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Coming soon...

1. Strengthen the future-oriented thinking of the master planning
2. Examine potential development trajectories for Helsinki’s operating environment up to 2075.
3. Identify land-use measures at the master plan level to strengthen resilience.
4. Develop and test a land-use simulation model to depict alternative land-use scenarios.



JÄTKÄSAARI 2009



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JÄTKÄSAARI 2022



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JÄTKÄSAARI 2030



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GIS in urban planning and decision-making

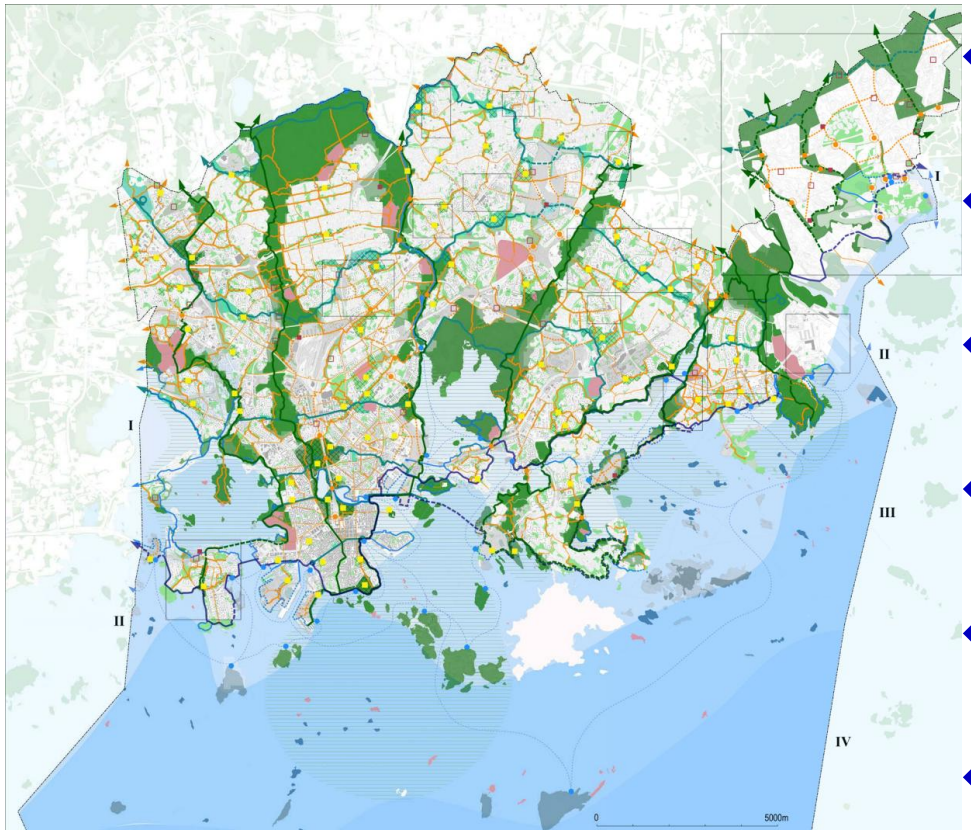
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Challenges:

- Monitoring change over time
- Better coordination across scales & agencies
- Integrating heterogeneous spatial data layers
- Quantifying
- Detecting spatial patterns

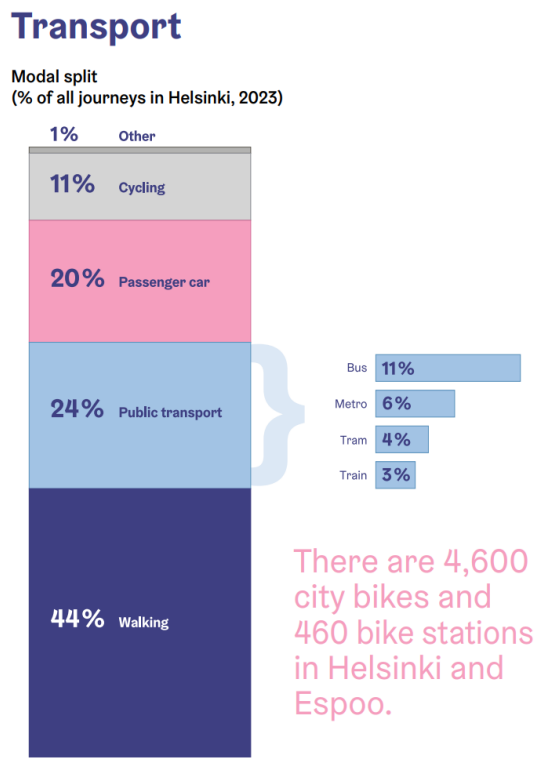
GIS Advantages:

- Overlay
- Shared GIS base layers
- spatial joins and attribute linking
- Isochrone / network / cumulative opportunity analysis
- Spatial statistics



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GIS used for: route planning, accessibility analysis, travel time modeling, evaluating impacts on land use and modal shift



There are 4,600 city bikes and 460 bike stations in Helsinki and Espoo.



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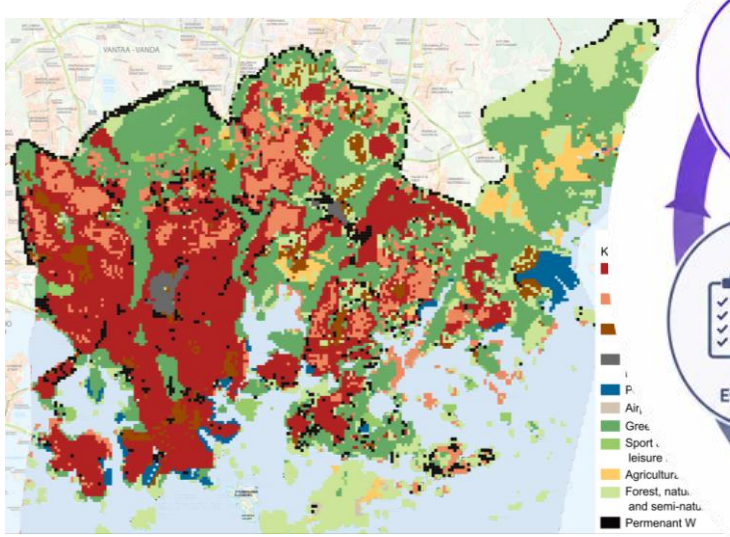

KRUUNUSILLAT
Only for trams,
cyclists & pedestrians
Helsinki
Helsingfors

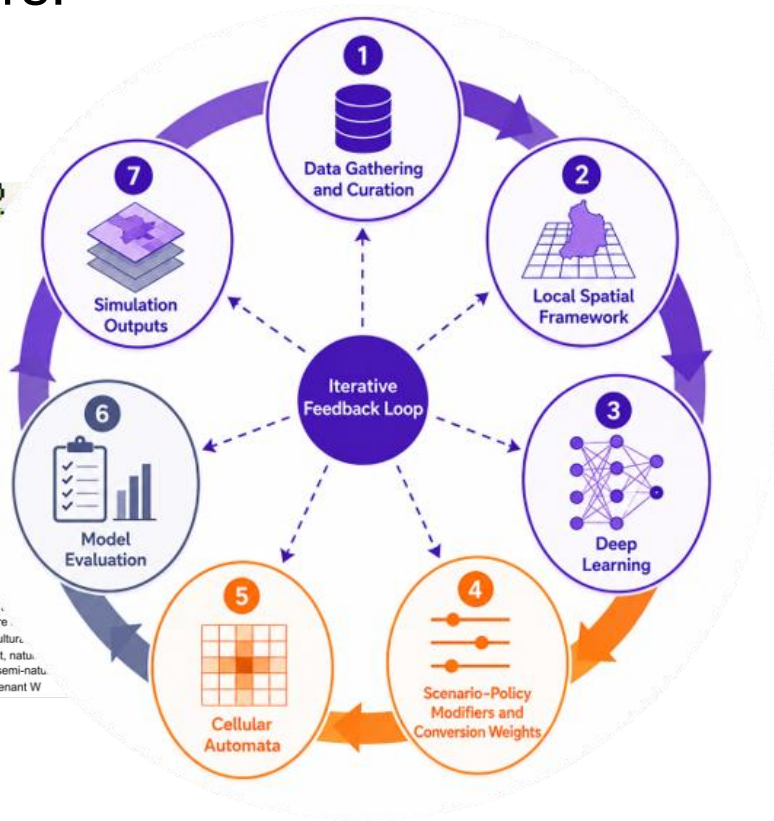
GIS and Land use simulation:

Simulation does not mean predicting the future; instead, it involves navigating the future.

Land use simulation model developed using GIS and Deep learning algorithms.
It learns from historical land use data in following years:

- 1990
- 2000
- 2006
- 2012
- 2018
- 2023



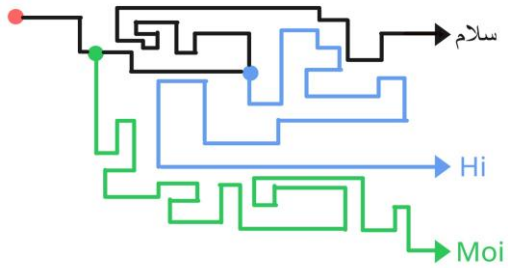


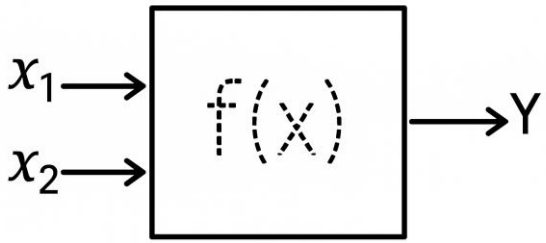
Unlocking Potentials of AI in Planning

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AI ≠ LLMs

Think of it as a brain, just like yours





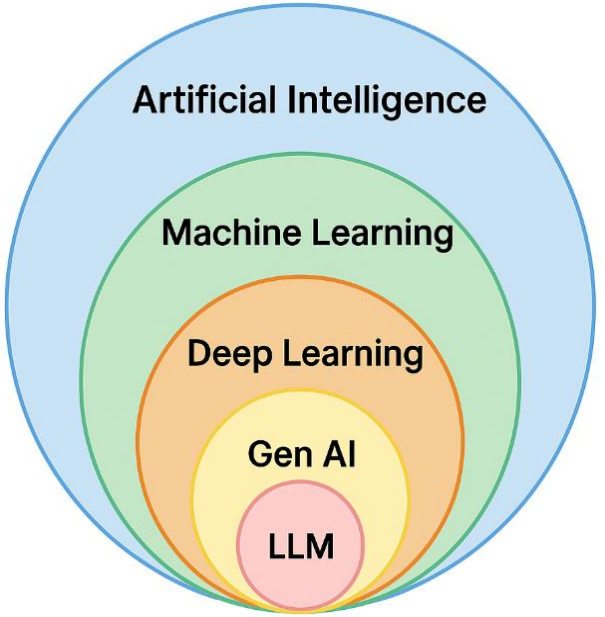
$f(x_1 = ?, x_2 = ?) =$ سلام
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$f(x_1 = ?, x_2 = ?) =$ Moi


Yoshua Bengio


Geoffrey Hinton

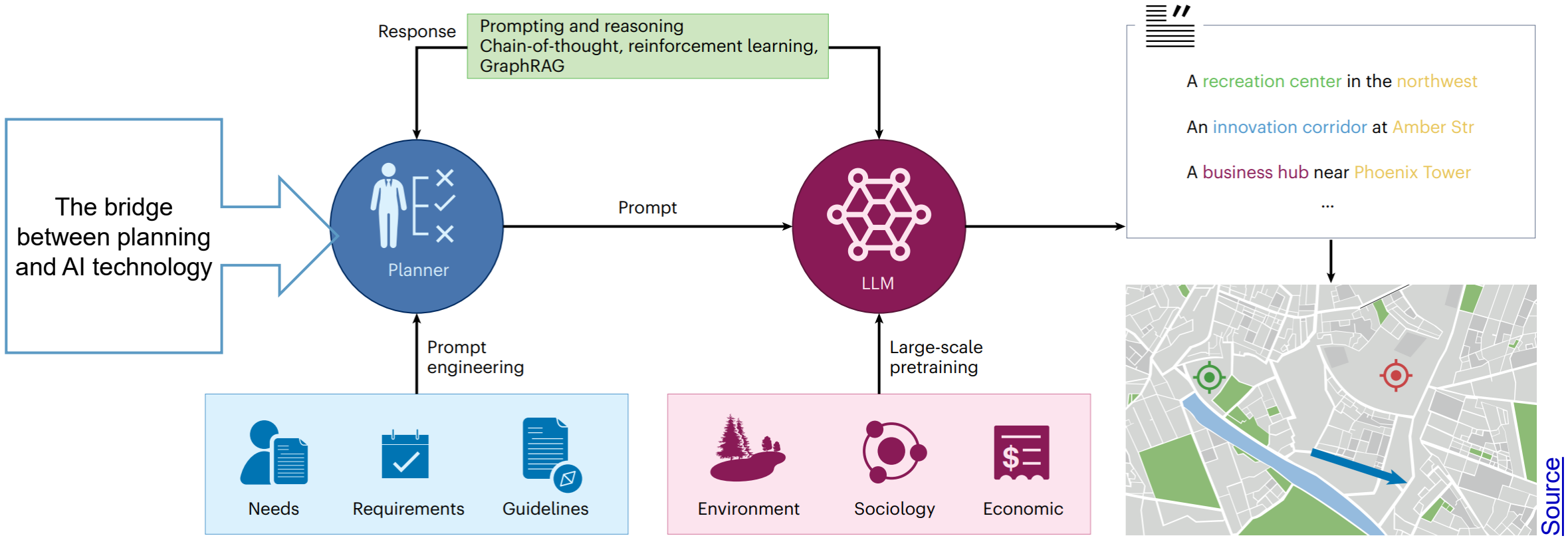

Yann LeCun

- **1943 –** Neuroverkot
- **1957–1998 –** Algritmien kehitys
- **2006 –** Syväoppiminen
- **2020 –** Suuret kielimallit ja muut mallit



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The role of urban planner



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Promoting the use of artificial intelligence

- We compile and specify the use cases and expected benefits of artificial intelligence.
- We will guide our colleagues to suitable technical, information security and management support.
- We support AI experiments, the introduction of Copilot and competence development.
- We highlight initiatives that require decisions, resources or funding.
- We maintain an overall picture of AI activities in our unit.



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12.8.2025

Current projects and ideas

01

Päätöskaivuri

Clarifies and condenses decisions and makes it easier to find information.

DECISION INFORMATION

02

Vaatimushallinta-apuri

Helps to find and compare urban planning requirements.

REQUIREMENTS

03

Rahoitushakkuri

Compiles the search, comparison and assessment of the competitive situation of public funding applications.

FINANCING

04

Land-use simulation system

Collects foresight information to support zoning and decision-making.

The project is in the brainstorming phase.

FORESIGHT

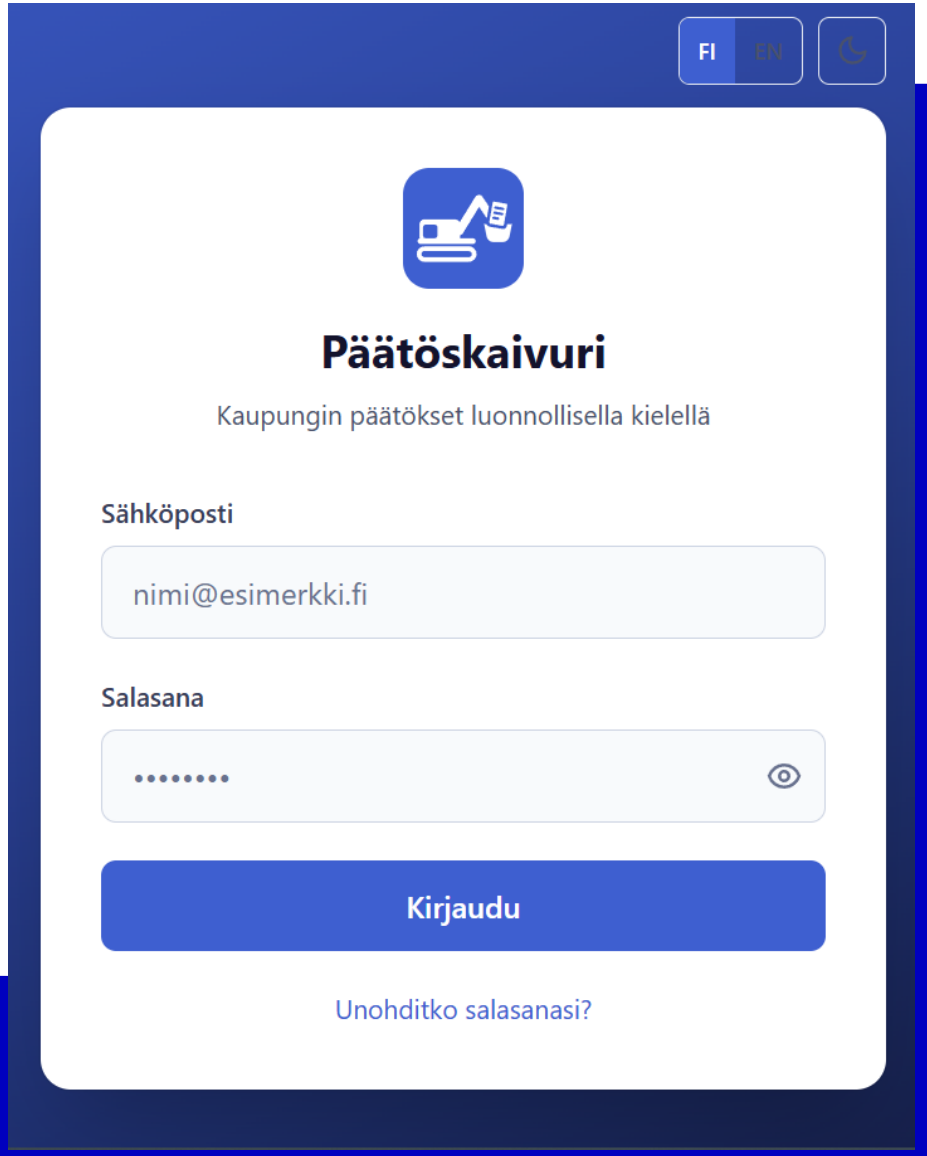
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Päätöskaivuri

is an AI-assisted search service of the City of Helsinki, which allows users to search for the city's decisions in natural language in Finnish or English and receive a condensed answer and links to the original sources.

- Päätöskaivuri is an MVP solution that works as a web application.
- At this point, it is for internal use only.
- 50 pilot users | 09/2026 – 02/2027 | 6 months

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Kuvat Roni Rekomaa